

Assignment #9: dfs, bfs, & dp

Updated 2107 GMT+8 Nov 19, 2024

2024 fall, Compiled by 同学的姓名、院系

说明:

1) 请把每个题目解题思路 (可选), 源码 Python, 或者 C++ (已经在 Codeforces/Openjudge 上 AC), 截图

图 (包含 Accepted), 填写到下面作业模版中 (推荐使用 typora <https://typoraio.cn>, 或者用 word)。AC 或者没有 AC, 都请标上每个题目大致花费时间。

2) 提交时候先提交 pdf 文件, 再把 md 或者 doc 文件上传到右侧 作业评论。Canvas 需要有同学清晰头像、提交文件有 pdf、"作业评论"区有上传的 md 或者 doc 附件。

3) 如果不能在截止前提交作业, 请写明原因。

1. 题目

18160: 最大连通域面积

dfs similar, <http://cs101.openjudge.cn/practice/18160>

思路:

代码:

```
m = int(input())
for _ in range(m):
    N,M = map(int,input().split())
    da = [['.' for _ in range(M+2)] for _ in range(N+2)]
    v = [[False for _ in range(M+2)] for _ in range(N+2)]

    for i in range(1,N+1):
        da[i][1:M+1] = input()

    step = [[-1,-1],[-1,0],[0,-1],[-1,1],[0,1],[1,1],[1,0],[1,-1]]

    ans = 0
    e = 0

    def search(v,da,x,y):
        global ans
        if da[x][y] == '.':
            return
        if v[x][y]:
            return
        ans += 1
        v[x][y] = True
        for j in range(8):
            nx = x+step[j][0]
            ny = y+step[j][1]
            search(v,da,nx,ny)

    for x in range(1,N+1):
        for y in range(1,M+1):
            search(v,da,x,y)
```

```
e = max(e,ans)
ans = 0

print(e)
```

代码运行截图 (至少包含有"Accepted")

状态: Accepted

源代码

```
m = int(input())
for _ in range(m):
    N,M = map(int,input().split())
    da = [['.' for _ in range(M+2)] for _ in range(N+2)]
    v = [[False for _ in range(M+2)] for _ in range(N+2)]

    for i in range(1,N+1):
        da[i][1:M+1] = input()

    step = [[-1,-1],[-1,0],[0,-1],[-1,1],[0,1],[1,1],[1,0],[1,-1]]

    ans = 0
    e = 0

    def search(v,da,x,y):
        global ans
        if da[x][y] == '.':
            return
        if v[x][y]:
            return
        ans += 1
        v[x][y] = True
        for j in range(8):
            nx = x+step[j][0]
            ny = y+step[j][1]
            search(v,da,nx,ny)

    for x in range(1,N+1):
        for y in range(1,M+1):
            search(v,da,x,y)
            e = max(e,ans)
            ans = 0

    print(e)
```

基本信息

#: 47347863
题目: 18160
提交人: 24n2400011491
内存: 3824kB
时间: 141ms
语言: Python3
提交时间: 2024-11-23 14:59:39

用时: 40min

19930: 寻宝

bfs, <http://cs101.openjudge.cn/practice/19930>

思路:

代码:

```
m,n = map(int,input().split())
mapp = [[2 for _ in range(n+2)] for _ in range(m+2)]
for i in range(1,m+1):
    mapp[i][1:n+1] = map(int,input().split())

move = [[1,0],[-1,0],[0,1],[0,-1]]

steps = []
def search(mapp,x,y,step):
    if mapp[x][y] == 2:
        return
    elif mapp[x][y] == 1:
        steps.append(step)
        return
    elif mapp[x][y] == 0:
```

```

        step +=1
        mapp[x][y] = 2
        for i in range(4):
            nx = x+move[i][0]
            ny = y+move[i][1]
            search(mapp,nx,ny,step)
        mapp[x][y] = 0

if mapp[1][1] == 1:
    print(0)
else:
    search(mapp, 1, 1, 0)
    if not steps:
        print('NO')
    else:
        print(min(steps))

```

代码运行截图 (至少包含有"Accepted")

用时: 24min

04123: 马走日

dfs, <http://cs101.openjudge.cn/practice/04123>

思路：先加两层保护圈，当搜索程序走到走过的位置上时判断此时的步数是否等于棋盘上最开始的空位数，如果等于，说明此时马已经走遍了棋盘，step+=1，但考虑到我是在最后马走完棋盘后再多一步才结算，结果会算成8倍，最后再除去就可以了。

代码：

```

move = [[1,2],[2,1],[2,-1],[-1,2],[-2,1],[1,-2],[-1,-2],[-2,-1]]

step = 0

def jump(mapp,x,y,k,w):
    global step
    if mapp[x][y] ==1:
        if k ==w:
            step +=1
            return
    else:
        mapp[x][y] = 1
        for i in range(8):
            nx = x +move[i][0]
            ny = y +move[i][1]
            jump(mapp,nx,ny,k+1,w)
        mapp[x][y] = 0

t = int(input())
for _ in range(t):
    n,m,x,y = map(int,input().split())
    mapp = [[1 for _ in range(m+4)] for _ in range(n+4)]
    for i in range(2,n+2):
        mapp[i][2:m+2] = [0 for _ in range(m)]
    step = 0
    jump(mapp,x+2,y+2,0,n*m)

```

```
print(step//8)
```

代码运行截图 (至少包含有"Accepted")

状态: Accepted

源代码

```
move = [[1,2],[2,1],[2,-1],[-1,2],[-2,1],[1,-2],[-1,-2],[-2,-1]]

step = 0

def jump(mapp,x,y,k,w):
    global step
    if mapp[x][y] == 1:
        if k == w:
            step += 1
        return
    else:
        mapp[x][y] = 1
        for i in range(8):
            nx = x + move[i][0]
            ny = y + move[i][1]
            jump(mapp,nx,ny,k+1,w)
        mapp[x][y] = 0

t = int(input())
for _ in range(t):
    n,m,x,y = map(int,input().split())
    mapp = [[1 for _ in range(m+4)] for _ in range(n+4)]
    for i in range(2,n+2):
        mapp[i][2:m+2] = [0 for _ in range(m)]
    step = 0
    jump(mapp,x+2,y+2,0,n*m)
    print(step//8)
```

基本信息

#: 47350528
题目: 04123
提交人: 24n2400011491
内存: 3620kB
时间: 4293ms
语言: Python3
提交时间: 2024-11-23 16:25:36

sy316: 矩阵最大权值路径

dfs, <https://sunnywhy.com/sfbj/8/1/316>

思路: 按照深度搜索写的代码, 但这里最大的坑点是列表在迭代的时候有个浅拷贝的问题。

代码:

```
move = [[0,1],[1,0],[0,-1],[-1,0]]

n,m = map(int,input().split())
matrix = [[-101 for i in range(m+2)] for j in range(n+2)]
for i in range(1,n+1):
    matrix[i][1:m+1] = map(int,input().split())
key = float('-inf')
ans = []

def search(matrix,x,y,k,steps):
    global key
    global ans
    if matrix[x][y] == -101:
        return
    steps.append([x, y])
    k += matrix[x][y]
    if x == n and y == m:
        if k > key:
            key = k
            ans = steps.copy()
        return
    a = matrix[x][y]
    matrix[x][y] = -101
    for i in range(4):
        nx = x+move[i][0]
```

```

        ny = y+move[i][1]
        search(matrix,nx,ny,k,steps[:])
        matrix[x][y] = a

search(matrix,1,1,0,[])
for j in ans :
    print(*j)

```

代码运行截图 (至少包含有"Accepted")

```

1  move = [[0,1],[1,0],[0,-1],[-1,0]]
2
3  n,m = map(int,input().split())
4  matrix = [[-101 for i in range(m+2)] for j in range(n+2)]
5  for i in range(1,n+1):
6      matrix[i][1:m+1] = map(int,input().split())
7  key = float('-inf')
8  ans = []
9
10 def search(matrix,x,y,k,steps):
11     global key
12     global ans
13     if matrix[x][y] == -101:
14         return
15     steps.append([x, y])
16     k += matrix[x][y]
17     if x == n and y == m:
18         if k > key :
19             key = k
20             ans = steps.copy()
21         return
22     a = matrix[x][y]
23     matrix[x][y] = -101
24     for i in range(4):
25         nx = x+move[i][0]
26         ny = y+move[i][1]

```

测试输入

提交结果

历史提交

完美通过

[查看题解](#)

100% 数据通过测试

运行时长: 0 ms

LeetCode62.不同路径

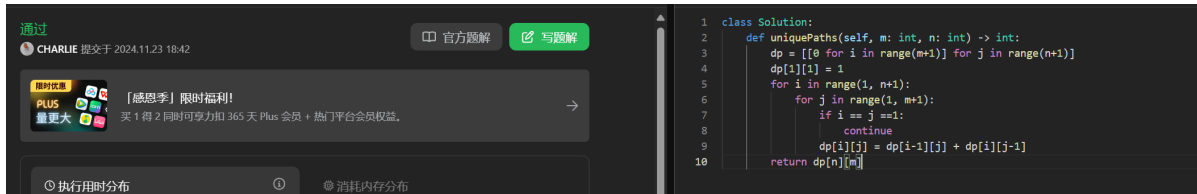
dp, <https://leetcode.cn/problems/unique-paths/>

思路:

代码:

```
class Solution:
    def uniquePaths(self, m: int, n: int) -> int:
        dp = [[0 for i in range(m+1)] for j in range(n+1)]
        dp[1][1] = 1
        for i in range(1, n+1):
            for j in range(1, m+1):
                if i == j == 1:
                    continue
                dp[i][j] = dp[i-1][j] + dp[i][j-1]
        return dp[n][m]
```

代码运行截图 (至少包含有"Accepted")



sy358: 受到祝福的平方

dfs, dp, <https://sunnywhy.com/sfbj/8/3/539>

思路：做这道题的时候我一直在思考贪心策略，最后才发现没法解决的反例，看了题解才知道要用dfs。

代码：

```
squares = []
i = 1
while i**2 <= 10**9:
    squares.append(i**2)
    i += 1

a = list(map(int, str(input()))))

def dfs(k, a):
    if k == len(a):
        return True

    num = 0
    for i in range(k, len(a)):
        num = num*10 + a[i]
        if num in squares:
            if dfs(i+1, a):
                return True
    return False

if dfs(0, a):
    print('Yes')
else:
    print('No')
```

代码运行截图 (至少包含有"Accepted")

```

1 squares = []
2 i = 1
3 while i**2 <=10**9:
4     squares.append(i**2)
5     i +=1
6
7 a = list(map(int, str(input())))
8
9 def dfs(k, a):
10     if k == len(a):
11         return True
12
13     num = 0
14     for i in range(k, len(a)):
15         num = num*10 + a[i]
16         if num in squares:
17             if dfs(i+1, a):
18                 return True
19     return False
20
21 if dfs(0, a):
22     print('Yes')
23 else:
24     print('No')
25

```

测试输入

提交结果

历史提交

完美通过

100% 数据通过测试

运行时长: 0 ms

2. 学习总结和收获

如果作业题目简单，有否额外练习题目，比如：OJ“计概 2024fall”每日选做、CF、LeetCode、洛谷等网站题目。

期中考试终于结束了，我这周抽空把每周发的讲义看了一下，感觉收获很大。我发现现在越来越多的算法是自己很难凭空想出来的，只有接触过一些相关的例子才好举一反三。

感觉dp和dfs的难点很奇怪，发现问题可以这样解决后代码并不难写，但难在发现问题适合采用这样的解法。